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The importance of spatial scale for the conservation of tidal flat macrobenthos: An example from New South Wales, Australia

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ABSTRACT

Planners of Marine Protected Areas (MPAs) commonly use maps of habitat types when choosing areas to conserve. This assumes that habitats are homogeneous, and therefore, that any area of habitat will represent the full spectrum of ecological diversity within that habitat. Here, we report that macrobenthic assemblages in tidal flat habitats were spatially heterogeneous in terms of beta diversity (taxonomic turnover), abundance, taxonomic richness and Shannon–Wiener H′ Diversity. Importantly, the patterns of heterogeneity were scale dependent for the three spatial scales we examined; plots (20 m), sites (100s of m) and estuaries (<30 km). The three estuaries in the study were compositionally similar as they shared the same dominant taxa, although one estuary had significantly more taxa and a higher abundance of macrobenthos. Assemblages within tidal flats differed at scales of 100s of m for all ecological measures. Most notably, beta diversity was highest at this scale. Assemblages were relatively more homogeneous at the 20-m scale. These findings highlight the value of examining more than one ecological measure and estimating magnitude of effects across a variety of scales. This work presents two important considerations for MPAs. First, although tidal flats in different estuaries are compositionally similar for dominant taxa, rarer taxa and high heterogeneity in abundance should influence the choice and number of tidal flats in MPAs. Second, strong compositional heterogeneity within individual tidal flats implies that conservation of whole habitat, rather than sections of a tidal flat, is essential if this habitat type is to be used to represent taxonomic diversity.
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1. Introduction

The design of MPAs is an emerging science that utilizes experiences from the design of terrestrial conservation areas, but also requires new approaches that address the unique challenges of the marine environment (Pressey and Mc Neill, 1996). Historically, MPAs have been designed in an ad hoc or opportunistic fashion (Pressey, 1994; Avery, 2003). This is due in part to limited knowledge of the patterns of biological distributions from the marine environment (Boersma and Parrish, 1999; Stewart et al., 2003). Conservation efforts are also usually heavily compromised due to perceived socio-economic impacts by a range of stakeholders (Roberts and Hawkins, 2000). Consequently, conservation efforts may be

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inefficient, inadequate or even fail to achieve set goals for reserve systems (Pressey, 1995; Stewart et al., 2003). To avoid this, many MPAs are now mandated to be designed according to guidelines of comprehensiveness, adequateness and representativeness (CAR) (Burgman and Lindenmayer, 1998; Lubchenco et al., 2003). CAR reserve systems must contain multiple representative samples of all ecosystems, habitats and associated taxa within bioregions (ANZECC TFMPA, 1998). Bioregions in Australian marine waters are areas that delineate ecological characteristics distinct from other areas in Australia, and are usually at scales of 100–1000s of km (ANZECC TFMPA, 1999).

A precursor to establishing a CAR system of MPAs is mapping the different habitat types within each bioregion (Pressey and Ferrier, 1995; Jordan et al., 2005), which function as surrogates for ecological diversity (Ward et al., 1999; Stewart et al., 2003). During planning, these maps are used to choose habitat reserves. However, numerous critiques of MPA design state that an important limitation to this approach is a lack of systematically surveyed biological data (Ward et al., 1999; Avery, 2001; Banks and Skilleter, 2002; Ponder et al., 2002). In particular, there is limited information on the spatial scales at which faunal assemblages change in taxonomic composition within a habitat type. It is well established that taxa do turn over between assemblages along a distance or environmental gradient (Jablonski and Sepkoski, 1996), which is best described by the term beta diversity (Pielou, 1974; Clarke and Cranme, 1997; Gray, 2000). There are diverse applications for the concept of beta diversity (Gering and Crist, 2002; Ricotta et al., 2002; Koleff et al., 2003), but here it is used according to early definitions (Pielou, 1974) and a review by Gray (2000), where beta diversity can describe taxonomic turnover in assemblages at various spatial scales within a habitat. Beta diversity is an important ecological measure for the siting of Marine Protected Areas that aim to represent the full range of taxa in a region. Conservation planners need to consider beta diversity for two reasons; first, if beta diversity is low at the scale of bioregion, i.e. assemblages consistently contain the same taxa in all habitat locations, then any location has the potential to be a comprehensive and representative area for conservation. If, however, beta diversity is high at the scale of bioregion, then multiple habitat locations that encompass this beta diversity will be required to achieve comprehensiveness and representativeness. Second, at smaller scales, interpreting habitat as an indicator of biological homogeneity can be misleading as many taxa exhibit patchy distributions within what appears to be homogeneous habitat (Thrush, 1991; Stevens and Connolly, 2004). This assumption is rarely tested for conservation purposes despite habitat heterogeneity being documented in the scientific literature (Thrush, 1991; Hacking, 1998; Dittman, 2000; Curley et al., 2002; Stevens and Connolly, 2004; Rodrigues et al., 2006). Heterogeneity may be an important defining characteristic of many habitats (Holt et al., 2004; Tews et al., 2004), but it is rarely known what scales of habitat and associated heterogeneity are required to represent or adequately "maintain the status of" assemblages (Constable, 1999). Consequently, if a portion of habitat is protected and scales of beta diversity are greater than the reserve area, there is a risk that the goal of representing species and ecological heterogeneity in the reserve will not be achieved. If habitats are to be mapped as surrogates for ecological diversity, then spatial scales of ecological heterogeneity or the processes that drive habitat heterogeneity need to be known to ensure that the full range of taxa are adequately represented in reserves.

Estuarine tidal flats are one habitat used as a planning unit and surrogate for taxonomic diversity in Australian MPAs (NSW Marine Parks Authority, 2001). This habitat is an important transition zone, with ecological processes that functionally link terrestrial, freshwater and marine ecosystems (Wall et al., 2001; Gray, 2002). For example, tidal flats are a principal energy link between primary production and consumers such as birds and fish (Loneragan and Bunn, 1999; McLusky, 1999; Levin et al., 2001). Therefore, it is important to include tidal flats in a system of MPAs.

Despite considerable international research on the ecology of estuarine tidal flats (Thrush and Warwick, 1997; Constable, 1999), there have been relatively few Australian studies (Fairweather and Quinn, 1995). Existing studies report small and intermediate scale spatial patterns of assemblages or taxa in tidal flats (see reviews by Thrush (1991), Constable (1999) and Gray (2002)), but they are rarely presented in a context that can contribute to conservation planning. There are some notable exceptions, and in particular, Edgar et al. (2000) identified discrete biological assemblages associated with up to nine different morphological types of estuaries (e.g. barred estuaries, drowned river valleys, etc.), as well as considerable biological variation at smaller scales within each estuarine type. However, taxonomic turnover (beta diversity) between estuaries of the same morphological type was not determined.

The aims of this research were to determine which of three spatial scales (20 m, 100s of m and <30 km) of ecological heterogeneity are most important for conservation planning of tidal flat habitat on the South Coast of NSW. We measured beta diversity as well as three other ecological measures: abundance, taxonomic richness and Shannon Wiener Diversity (H′), to determine if different measures showed the same patterns. The results should be relevant to spatial conservation planning in MPAs by answering practical questions: (a) Does a single tidal flat provide a comprehensive and representative reflection of other tidal flats in the region? (b) Do we need to include whole tidal flats in reserves to represent all infaunal macrobenthos? (c) What further research is necessary in order to optimize the selection of tidal flats in MPAs?

2. Methods

2.1. *Study sites*

Three similar, permanently open, barred estuaries on the south coast of New South Wales (NSW), Australia were chosen as study sites (Fig. 1) within a larger temporal study (Winberg, in preparation). The first year of data was used for this study in order to describe spatial ecological heterogeneity. In each estuary, the first tidal flat upstream – within 1 km of the estuarine mouth – was sampled. One of the tidal flats, Currambene Creek (35° 01′ S, 150° 40′ E), was proposed as a Sanctuary Zone (no take zone) in the Jervis Bay Marine Park (JBMP). The two other estuaries were at Sussex Inlet (35° 10′

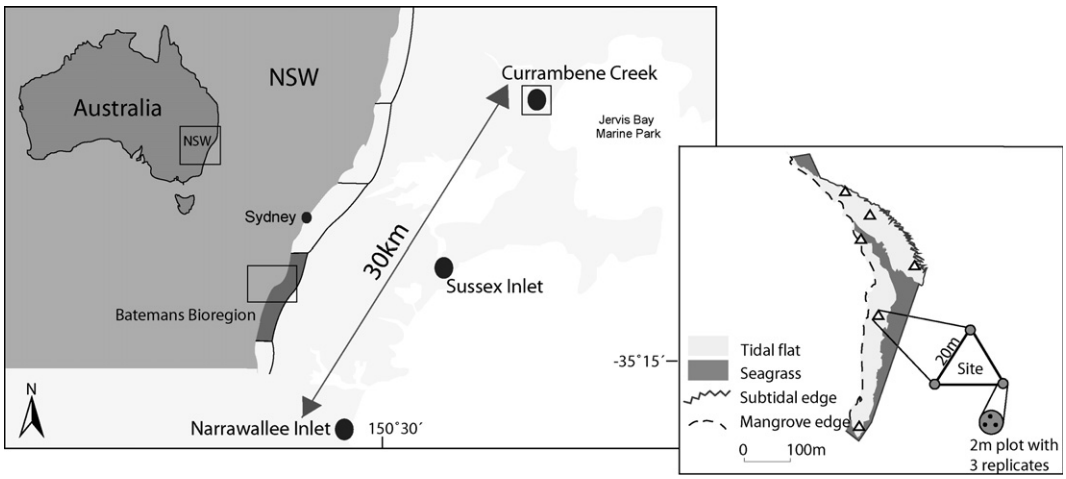

Fig. 1 – Study locations at three estuaries in the Bateman's Bioregion of NSW, Australia. The first tidal flat upstream from the estuary mouth was the location studied, and the general sampling design is illustrated for the Currambene Creek tidal flat. Three spatial scales were examined: plots (20 m), sites (100s of m) and estuaries (<30 km).

S, 150° 35′ E) and Narrawallee Inlet (35° 18′ S, 150° 28′ E). All of these estuaries are within 30 km of each other in the Bateman's Bioregion.

The sampled tidal flats in each estuary were directly adjacent to fully marine waters with similar salinity regimes and both the estuaries and tidal flats were morphologically similar. Each tidal flat consisted of extensive unvegetated, sandy habitat bordered sub-tidally by the seagrass *Zostera capricorni* or sub-tidal soft sediments, while the terrestrial border was fringed with mangroves and saltmarsh. The clay and silt content never exceeded 10%, although some sites appeared muddy (Winberg, unpublished data). The surfaces of the flats showed evidence of intense bioturbation, with burrow openings of Callianassid species and various polychaetes, as well as pits, ranging in size from a few centimetres to >1 m, made by benthic predators such as rays. At low tide the sediment was further disturbed by soldier crabs (*Mictyris* sp.) and amphipods, as well as by people pumping up bait from the sediment.

2.2. *Sampling design*

Each of the three estuaries was sampled on two occasions between July and October, 2002. On each sampling day three 20 m triangular sites were haphazardly selected at least 100 m from each other within each tidal flat (Fig. 1). At each apex, 2 m^2 plots were randomly sampled by taking three sediment cores within them. Hence, 27 cores were taken from each tidal flat on two occasions giving a total of 54 cores per tidal flat, and 162 cores in total.

Cores were 15 cm in diameter and 25 cm deep. Sediment from the cores was sieved in situ through 1.2 mm mesh bags, and frozen on return to the lab. The large mesh size relative to other studies that involve macrofaunal sampling was due to a number of sites containing sandy coarse sediments and the large volume of sediment sampled in an attempt to sample deep burrowing fauna. Samples were later defrosted and stained with Biebrich Scarlet in a 5% formol solution for at least 2 days. Taxa were identified to the lowest taxonomic level possible, usually species, using a dissecting microscope. The species level was chosen as it is suggested that lower taxonomic resolution should be used with caution (Vanderklift et al., 1998), especially as the suite of taxa sampled had not been examined previously at these scales. Rarer, juvenile or difficult taxa were identified to the genus or family level.

2.3. *Analysis of data*

Data were visualized in species-sample accumulation curves (999 permutations) using the PRIMER 5 software package (Plymouth Marine Laboratories, Clarke, 1993) to determine if the majority of species present had been sampled. Beta diversity was determined using multivariate data of taxonomic composition and calculating Bray-Curtis similarities with the PRIMER 5 software package. As this paper intends to contribute to the representativeness of taxa in MPAs, multivariate analyses were done using presence/absence data to give equal weight to rarer taxa (Clarke, 1993). Data were visualized in non-metric Multi-Dimensional Scaling (MDS) plots. We tested for assemblage differences at three spatial scales in a nested hierarchy on two occasions using the PERMANOVA software package (Anderson, 2001, 2005; McArdle and Anderson, 2001). As replication at the smallest scale of sampling (cores per plot) was 3 ($n = 3$), we used Monte Carlo Probability values in the analysis (Anderson, 2005). Pair-wise a posteriori comparisons were performed within the PERMANOVA software package using a multivariate version of the *t*-test (Anderson, 2005). This method was used as a measure of beta diversity as it determines whether there are significant differences in assemblage composition at the scales sampled.

To test for differences in abundance, taxonomic richness and log base e Shannon–Wiener H′ Diversity (diversity H′), a nested four factor ANOVA was used. Assumptions of ANOVA were examined prior to proceeding with analyses. Normality was assessed visually in plots of means against variances (Quinn and Keough, 2002). Cochran's test was used to test for homogeneity of variance, and abundance data were transformed ($(x + 1)^{1/2}$) to achieve homogeneity

(Quinn and Keough, 2002). Where groups differed significantly ($p < 0.05$), a posteriori comparisons among means were done with Student–Newman–Keuls tests. All ANOVAs were performed with GMAV 5 software (University of Sydney, Sydney, Australia).

For all the dependent variables (ecological measures) above, the relative contributions of the different spatial scales to overall heterogeneity was determined by partitioning the variation between the nested factors (scales) manually, according to methods in Graham and Edwards (2001). The term ω was used for the percentage contribution to overall variation as is consistent with Graham and Edwards (2001). For example, if assemblages differed significantly from each other at both the scales of estuaries and sites, ω indicates whether one of those scales exhibited greater heterogeneity than the other.

To determine the relative contributions to variation and the proportion of variation between scales of sites and plots were consistent among estuaries, the PERMANOVA and ANOVA procedures were repeated using data from each estuary alone (therefore testing scales of sites and plots only). As the design was fully balanced, we used the Sums of Squares for each of the nested factors in the separate estuary analyses to determine the proportional contribution to overall variation at the respective scales in the full data set. Additionally, a measure of Dispersion and an Index of Multivariate Dispersion was obtained for multivariate data only, using the MVDisp procedure in the PRIMER 5 software package.

3. Results

A total of 4468 animals representing 54 taxa were found in cores taken from the three estuaries (Table 1). Cumulative samples from each estuary were representative as species accumulation curves approached an asymptote for each estuary (Fig. 2). Subsequent data from the following 2 years also indicated that the majority of species had been sampled (Winberg, in preparation). 70–84% of all animals in each estuary were attributed to four common taxa: two bivalves *Mysella anomala* and *Eumarcia fumigata*, and two crustracean taxa, *Urohaustorius metungi* and *Mictyris* species. The factor "Time" was eliminated and its' variance pooled in all analyses as there was no significant temporal variation ($p > 0.25$). Only spatial components are reported below.

3.1. *Estuaries*

Comparison of assemblages in different estuaries indicated that beta diversity was insignificant at the scale of estuaries ($p = 0.20$, $F = 1.4$) (Fig. 3a and c, Table 2), and therefore that a single tidal flat could be considered representative of the assemblages across tidal flats in the three estuaries.

In terms of abundance and taxonomic richness, Currambene Creek macrobenthos were nearly three fold more abundant ($p = 0.01$, $F = 6.33$) and contained twice as many taxa ($p = 0.01$, $F = 6.08$) compared to Sussex and Narrawallee Inlets (Fig. 3b, Table 2). The scale of estuaries contributed 34% to total variation for both abundance and taxonomic richness. Currambene Creek also had higher diversity H′ than the other two estuaries ($p = 0.06$, $F = 3.52$), though the contribution to assemblage variation was lower (17%) than for abundance and taxonomic richness.

3.2. *Sites*

Macrofaunal assemblages differed at the scale of sites (100s of m) for all ecological measures across all estuaries (Table 2). Notably, a strong pattern of site aggregation was evident in each of the estuaries and was clearly visualized in MDS plots (Fig. 4, Table 2). This site aggregation was reflected in the highest compositional variation (39%) of assemblages at all scales (Fig. 3c). Consequently, beta diversity (taxonomic turnover) across sites at each estuary was high compared to beta diversity across estuaries. Variation in diversity H′ (27%) was also higher at the scale of sites than at the scale of estuaries. However, abundance and taxonomic richness contributed to between 30% and 35% of total variation at the scale of sites and estuaries.

The relative contribution to variation at the scale of sites differed between estuaries and ecological measure (Fig. 3c). Currambene Creek drove variation in abundance and taxonomic richness at this scale, while there was a more equal contribution to variation between sites for Shannon H′ Diversity and beta diversity from each of the estuaries. The measure of dispersion indicated that the relative magnitude of variation in composition was similar in Narrawallee Inlet (dispersion = 1.112) and Sussex Inlet (dispersion = 1.052, IMD Narrawallee = 0.065), and that dispersion in Currambene Creek was lowest (dispersion = 0.835, IMD Sussex = 0.222, IMD Narrawallee = 0.272). However, Currambene Creek still contributed to overall site variation with a magnitude comparable to that of the other estuaries (Fig. 3c).

3.3. *Smaller scale variation*

There were significant differences in all variables at the scale of plots; however, the contribution to overall variation was relatively low for abundance, taxonomic richness and beta diversity (Fig. 3c). The variation of diversity H′ was similar at distances of 20 m (plots) to distances of 100s of m (sites).

The relative contribution to variation at the scale of plots differed between estuaries and ecological measure as for the scale of sites. Again, Currambene Creek drove variation in abundance and taxonomic richness at this scale, but Narrawallee Inlet also contributed to variation in abundance, while Sussex Inlet contributed to variation in taxonomic richness. The scale of sites contributed proportionally more variation than the scale of plots for all ecological measures in all estuaries, except for abundance in Narrawallee Inlet and taxonomic richness in Sussex Inlet.

4. Discussion

This study found low beta diversity among different tidal flats within a 30 km scale on the NSW south coast. There was little taxonomic turnover between assemblages, and the same taxa dominated all tidal flats. This is consistent with Gray (2002) who contends that regional diversity drives local diversity in soft sediment habitats. Despite this indication of low regional beta diversity, taxonomic richness was close to

Table 1 – Taxa ranked by mean abundance per sample core in each location

Taxon	Ph	Currambene				Sussex				Narrawallee			
		Rank	Mean	SD	%	Rank	Mean	SD	%	Rank	Mean	SD	%
Galeommatidae (*Mysella anomala*)	B	1	24.54	26.47	43.8	2	2.81	5.41	20.1	1	4.76	10.25	31.9
Urohaustoriidae (*Urohaustorius metungi*)	A	2	13.43	13.75	26.2	1	9.65	7.43	51.7	2	3.02	4.58	18.9
Mictyridae (*Mictyris* sp.)	C	3	3.41	2.47	6.9	4	0.81	1.34	6.0	3	1.50	1.84	10.5
Veneridae (*Eumarcia fumigata*)	B	4	1.22	1.34	2.8	3	1.13	2.24	6.3	4	1.17	1.38	8.5
Sigalionidae (*Sigalion ovigerum*)	P	5	0.94	1.37	2.1	8	0.19	0.40	1.2	12	0.15	0.35	1.1
Callianassidae (*Biffarius arenosa*)	C	6	0.83	1.50	2.4	12	0.04	0.19	0.2	7	0.44	0.94	3.2
Spionidae (*Carazziella victoriensis*)	P	7	0.57	1.29	1.0	13	0.02	0.10	0.2	19	0.02	0.10	0.1
Magelonidae (*Magelona dakini*)	P	8	0.56	0.82	1.3	5	0.28	0.70	2.0	6	0.46	0.81	3.3
Nemertea	N	9	0.48	0.79	1.0	12	0.04	0.13	0.2	13	0.13	0.43	0.9
Capitellidae (*Barantolla lepte*)	P	9	0.48	1.13	1.3	11	0.06	0.21	0.5				
Orbiniidae (*Scoloplos simplex*)	P	9	0.46	0.72	0.9	9	0.17	0.37	1.3	5	0.93	1.00	7.0
Psammobiidae (*Soletellina alba*)	B	9	0.46	0.79	0.8	10	0.09	0.29	0.6	18	0.04	0.19	0.2
Callianassidae (*Trypaea australiensis*)	C	10	0.44	0.83	0.8	12	0.04	0.13	0.2	15	0.09	0.24	0.7
Paracalliopiidae (*Paracalliope lowryi*)	A	11	0.41	1.10	1.2					10	0.31	0.65	2.2
Spionidae (*Prionospio yuriel*)	P	11	0.41	0.78	0.8								
Spionidae (*Prionospio aucklandia*)	P	12	0.37	0.77	1.1								
Nereididae (*Australonereis ehlersi*)	P	13	0.31	0.56	0.8	5	0.28	0.64	2.2	16	0.07	0.23	0.6
Capitellidae (*Capitella* sp.)	P	14	0.24	0.58	0.7	6	0.26	0.75	2.2	8	0.39	0.67	2.9
Spionidae (*Spio pacifica*)	P	14	0.24	0.52	0.7					18	0.04	0.19	0.3
Iravadiidae (*Nozeba topaziaca*)	G	15	0.20	0.56	0.3								
Nephtyidae (*Nephtys* sp.)	P	16	0.19	0.37	0.5	6	0.26	0.53	1.7	11	0.17	0.38	1.2
Capitellidae (*Mediomastus australiensis*)	P	16	0.19	0.31	0.5								
Phoxocephalidae (*Limnoporeia kingi*)	A	17	0.17	0.43	0.4					17	0.06	0.23	0.4
Tellinidae (*Tellina deltoidalis*)	B	18	0.11	0.31	0.2	10	0.09	0.24	0.4	18	0.04	0.13	0.2
Glyceridae (*Glycera tridactyla*)	P	19	0.09	0.28	0.3								
Naticidae	G	20	0.07	0.27	0.2	13	0.02	0.10	0.1	17	0.06	0.23	0.4
Buccinidae (*Nassarius* sp.)	G	20	0.07	0.33	0.2	12	0.04	0.13	0.2				
Lumbrinereidae (Lumbrineris sp. #1)	P	21	0.06	0.23	0.1								
Owenidae (*Owenia fusiformis*)	P	21	0.06	0.16	0.2								
Spionidae (*Pseudopolydora* sp.)	P	21	0.06	0.16	0.2								
Oligochaete	O	22	0.04	0.19	0.1					19	0.02	0.10	0.1
Spionidae (*Scolelepis* sp.)	P	22	0.04	0.19	0.1					19	0.02	0.10	0.1
Philinidae (*Philine* sp.)	G	22	0.04	0.13	0.1								
Gobiidae	F	22	0.04	0.13	0.1								
Thyasiridae	B	23	0.02	0.10	0.1								
Phyllodocidae (*Eumida* sp.)	P	23	0.02	0.10	0.1								
Phoronida	P	23	0.02	0.10	0.1								
Photidae (*Gammaropsis* sp.)	A	23	0.02	0.10	0.0					18	0.04	0.13	0.2
Penaeidae (*Penaeus plebejus*)	C	23	0.02	0.10	0.0					19	0.02	0.10	0.1
Sabellidae (*Desdemona aniara*)	P									9	0.33	0.92	2.5
Exoedicerotidae (*Exoediceros fossor*)	A					7	0.22	0.45	1.2	14	0.11	0.37	0.7
Hesionidae (*Heteropodarke* sp.)	P									16	0.07	0.18	0.5
Galeommatidae (sp. #3)	B									17	0.06	0.23	0.4
Cirolanidae (*Pseudolana towrae*)	I					12	0.04	0.19	0.2	18	0.04	0.13	0.2
Opheliidae (*Lobochesis* sp.)	P					13	0.02	0.10	0.1	19	0.02	0.10	0.1
Lucinidae (*Wallucina* sp.)	B									19	0.02	0.10	0.1
Cylichnidae (*Tornatina* sp.)	G									19	0.02	0.10	0.1
Actaeciidae	I									19	0.02	0.10	0.1
Bopyridae	I									19	0.02	0.10	0.1
Orbiniidae (*Leodamas johnstonei*)	P					9	0.17	0.28	0.8				
Mactridae	B					12	0.04	0.13	0.3				
Crangonidae	D					13	0.02	0.10	0.1				
Spionidae (*Scolelepis* sp. 2)	P					13	0.02	0.10	0.2				
Other						13	0.02	0.10	0.1				
Mean abundance/sample			51.3	36.5			16.8	9.9			14.6	12.6	
Mean species richness/sample			8.0	3.4			3.9	1.7			4.8	2.0	
Number of species/location			39				27				34		
Number of samples			54				54				54		
Total individuals			2771				790				907		

Standard deviation (SD) and percentage contribution to overall abundance (%) are shown. The Phyla or Class to which taxa belong are identified in column Ph. With the following key: A, amphipoda; B, bivalvia; C, crustacea; F, fish; G, gastropoda; I, isopoda; N, nemertea; O, oligochaete. Total values for each location are presented at the bottom of the table.

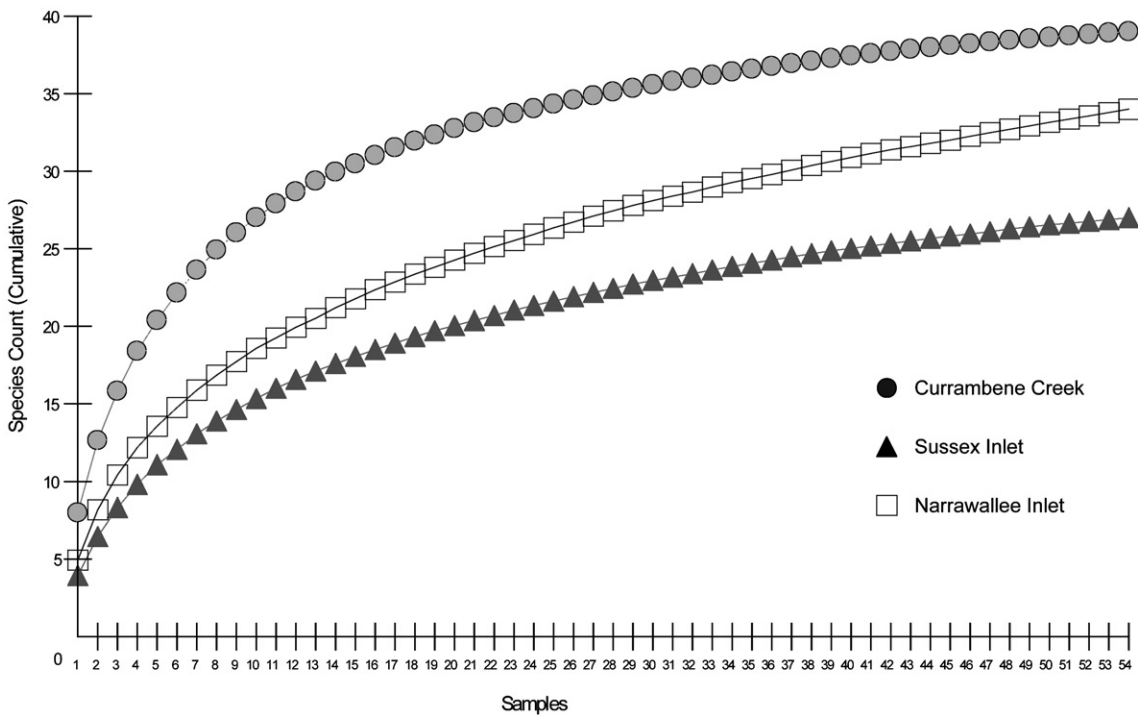

Fig. 2 – Species accumulation per sample curves for each of the tidal flats in each of the three estuaries Currambene Creek, Sussex Inlet and Narrawallee Inlet.

twofold higher in Currambene Creek. A full third of the taxa were found only in this tidal flat. The reason that this significantly higher proportion of taxa was not translated into significant change in beta diversity was due to the rarity of most of these additional taxa. Of course, some of these unique taxa may have been missed during sampling in other estuaries due to their rarity. However, alternative hypotheses may be that these taxa have truly small distributions or exist at their geographical range limits. Similarly, 11% and 12% of taxa found in Sussex Inlet and Narrawallee Inlet were unique to those tidal flats, respectively, although these differences did not translate into significant taxonomic richness differences between these two estuaries. Ultimately, conservation practitioners have to make choices regarding how many estuaries will provide representative MPAs, despite limitations in sampling rare taxa. If the goal of MPAs is to represent all taxa, then rare taxa cannot be discounted. Therefore, these findings suggest that a tidal flat may be broadly representative of other tidal flats within 30 km, in terms of dominant or common taxa, but the consequence of protecting only one tidal flat may be reduced representativeness of rare taxa. Multiple tidal flat reserves would increase the probability that representativeness is achieved.

A further consideration is that beta diversity will increase at some scale greater than the 30 km reported here. The coast of NSW encompasses the southern and northern limits for many tropical and temperate marine species, respectively (Turner et al., 2004), and other studies in the tropics and cold temperate regions as well as temperate estuaries in Europe contain different (although functionally similar) species (Dittman, 2000; Edgar and Barrett, 2002; Rodrigues et al., 2006). Another study (Ferrell et al., 1993) in the same region as this one identified compositional differences in estuarine assemblages of abundant fish species at scales of 100s of km. The 30 km scale presented here is a large increase in scale in terms of the majority of tidal flat research (Constable, 1999), but larger scale tidal flat research will be required to determine at what regional scales beta diversity increases significantly.

In addition to being the most taxonomically rich of the estuaries sampled, Currambene Creek also showed a near three fold higher density (abundance) of macrofauna than the other estuaries. Areas with high taxonomic richness and abundance (hotspots) have been suggested as criteria for nominating areas as candidates for conservation (Myers et al., 2000; Meir et al., 2004; Fox and Beckly, 2005). However, there is concern over whether such hotspots on their own can fulfil representation of MPAs (Fox and Beckly, 2005) or enhance recruitment processes and resilience of tidal flats (Carr et al., 2003; Gaines et al., 2003; Grantham et al., 2003).

Additionally, Currambene Creek drove the heterogeneity of abundance and taxonomic richness at the scales of sites (100s of m) and plots (20 m); which is possibly an effect of high abundance and taxonomic richness in this tidal flat (Holt et al., 2004). This was not translated into higher values of the two diversity indices, but is none the less a type of ecological heterogeneity important to consider for representative MPAs. Representation of tidal flat diversity may require that both heterogeneous and homogeneous habitat types are included in MPAs. For example, whether assemblages are more or less heterogeneous may affect subsequent resilience to different types of impacts (McClanahan et al., 2002). The implications and "benefits" associated with different degrees of such heterogeneity require investigation.

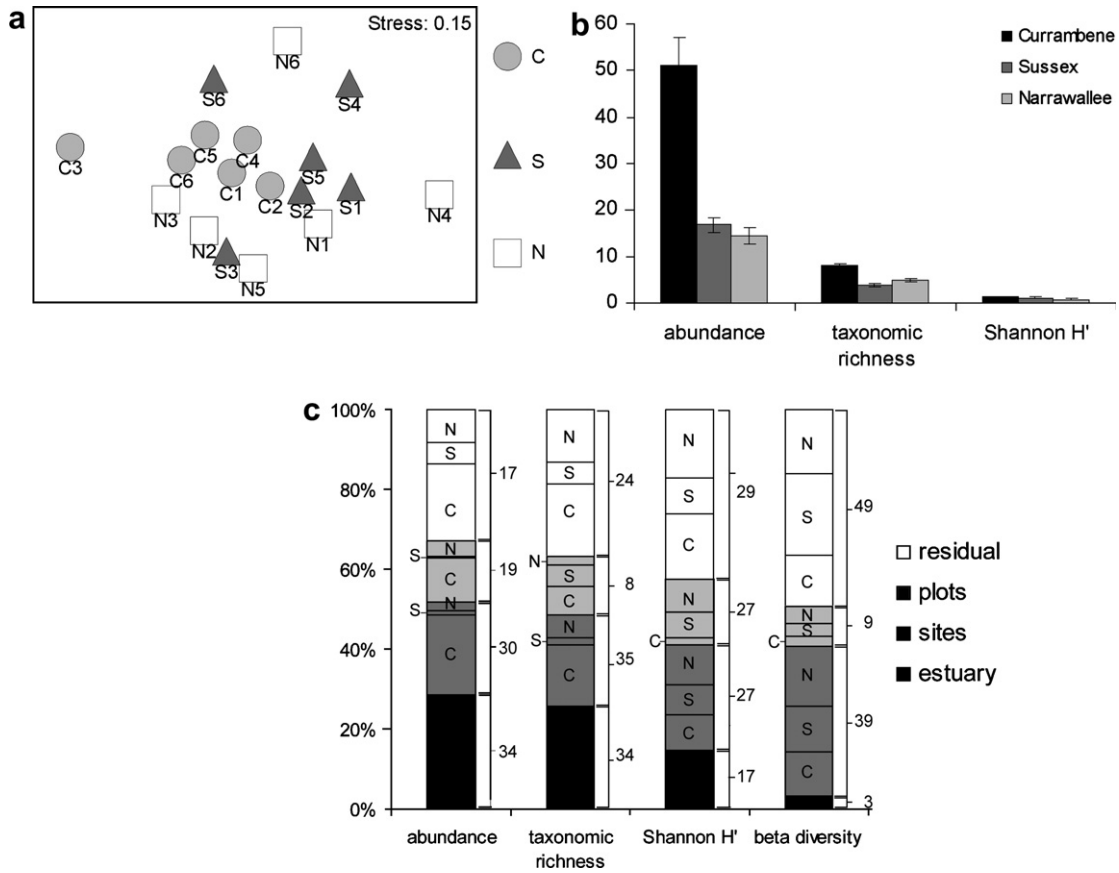

Fig. 3 – Summary of ANOVA and PERMANOVA results. (a) MDS plots of site averages using presence/absence multivariate data (beta diversity). (b) Abundance, Taxonomic Richness and Shannon H′ Diversity means in each of the estuaries. Standard error bars shown. (c) Magnitude of effects estimates of the different spatial scale contributions to the overall assemblage variation in all parameters for the three esturies Currambene Creek (C), Sussex Inlet (S) and Narrawallee Inlet (N).

Most notable at the scale of sites and common to all estuaries was a clear increase in heterogeneity in assemblage composition (beta diversity and diversity (H′)). Assemblage composition at one site in a tidal flat differed considerably from assemblage composition 100s of m away. Similar scales of species patchiness have been reported elsewhere (reviews in Peterson (1991), Thrush (1991), also Hewitt et al. (1997), and Dittman (2000)).

Predictive models of the processes causing assemblage patterns within tidal flats could improve our efficiency in choosing ideal locations for MPAs (Constable, 1999), particularly where tidal flats are large and/or where reserve selection may be compromised by other considerations. Patterns of assemblages are thought to be related to local environmental variables (Thrush, 1991; Posey et al., 1995; Hewitt et al., 1997; Edgar et al., 1999; Dittman, 2000; Edgar and Barrett, 2002). It is clear that the tidal flats we have examined exhibit heterogeneity in environmental factors at the scale of 100s of m; for example mud content and tidal inundation is highly variable among sites (Winberg, personal observation). If, however, a tidal flat extends over kilometres, scales at which environmental variables such as tidal height operate would also increase, resulting in increased scales of assemblage heterogeneity. One study of a much larger temperate tidal flat in Portugal (Rodrigues et al., 2006) supports this assertion. However, at this stage predictive models are limited (Constable, 1999), thus the results presented here illustrate the need to include whole habitat as reserve, rather than a portion of tidal flat, in order to encompass the full spectrum of ecological heterogeneity including all species.

Of concern is how long the observed patterns persist. Although shorter temporal seasonal changes were not detected in this study, which spanned four months from winter to spring, Silva et al. (2006) report seasonal heterogeneity in a temperate estuary in Portugal. Seasonal heterogeneity would intuitively also be important in other regions that experience dramatic seasonal changes such as temperatures below freezing or monsoonal rains. However, taking short term seasonal variation into account still allows for temporal persistence of spatial patterns over an annual scale. Assuming that patterns of assemblages at the scale of sites are driven by environmental factors, it follows that these patterns should be as persistent in time as the environmental variables themselves. Low levels of temporal variation persisted for the three years in which these locations have been sampled (Winberg, in preparation), and are in agreement with other research experience from temperate and cold temperate regions (Edgar, personal comment).

At the 20 m scale, reduced taxonomic turnover implies that assemblages were compositionally less heterogeneous

Table 2 – Summaries of PERMANOVA and ANOVA results comparing beta diversity, mean abundance, taxonomic richness and Shannon diveristy (H′)

	Factor	SS	df	MS	F	P	P(MC)	ω^2	%	*Transformation*
Abundance										$(x + 1)^{1/2}$
	Est	283	2	141.58	6.33	0.01	na	2.21	34	
	Si(Est)	336	15	22.38	4.71	<0.01		1.96	30	
	Pl(Si(Est))	171	36	4.75	4.4	<0.01		1.22	19	
	Residual	117	108	1.08				1.08	17	
	Total	906	161					6.47		
Taxonomic richness										None
	Est	494	2	247.04	6.08	0.01	na	3.82	34	
	Si(Est)	609	15	40.63	7.61	<0.01		3.92	35	
	Pl(Si(Est))	192	36	5.34	1.95	<0.01		0.87	8	
	Residual	296	108	2.74				2.74	24	
	Total	1592	161					11.35		
Shannon H′										None
	Est	7	2	3.41	3.52	0.06	na	0.05	17	
	Si(Est)	15	15	0.97	3.24	<0.01		0.07	27	
	Pl(Si(Est))	11	36	0.30	3.73	<0.01		0.07	27	
	Residual	9	108	0.08				0.08	29	
	Total	41	161					0.27		
Multivariate										p/a
	Est	29437	2	14718.70	1.3951	na	0.20	77.20	3	
	Si(Est)	158249	15	10549.92	5.6487		<0.01	964.69	39	
	Pl(Si(Est))	67236	36	1867.68	1.5642		<0.01	224.55	9	
	Residual	128956	108	1194.03				1194.03	49	
	Total	383878	161	2460.00				2460.47		

ω^2 and % values indicate the proportional contribution of each factor (weighting) to the total variation. P(MC) are Monte Carlo probability values calculated by PERMANOVA.

than at 100s of m. This is a similar finding to that of Kendall and Widdicombe (1999) who reported that shallow sub-tidal sediment assemblages were homogeneous at scales up to 50 m, with a trend to increased variability at 500 m. An explanation for the observed decrease in taxonomic turnover at this scale follows from the argument that heterogeneity at the 100s of m is driven by environmental factors that operate at that scale. Differences in tidal inundation, flow rates and influence from adjacent habitats would all be smaller at the 20 m scale than the 100s of m scale, and result in more homogeneity in structuring effects and the resulting assemblage. This scale would therefore represent alpha diversity of macrofauna in tidal flat habitat (Schluter and Ricklefs, 1993).

For the other ecological measures of abundance, taxonomic richness and Shannon H′ Diversity, heterogeneity also decreased between scales of 100s of m to 20 m with the exception of abundance in Narrawallee Inlet and taxonomic richness in Sussex Inlet. With general increased homogeneity at this scale as well as the trend for reserves in MPAs to be substantially larger than 10s of m (Halpern, 2003), heterogeneity of assemblages at 20 m will, in practice, be encompassed within a reserve. Some variation however, highlights the need for adequate replication at smaller scales in tidal flat research.

There is currently no consensus on an optimal set of ecological measures in order to properly compare studies from different regions, although it has been suggested that certain approaches are preferable to others (Fox and Beckly, 2005). Similar to other tidal flat research (Dittman, 2000; Edgar and Barrett, 2002; Rodrigues et al., 2006), we found that the application of a range of different measures of ecological diversity (abundance, taxonomic richness, diversity (H′) and beta diversity) described different aspects of ecological heterogeneity operating at different spatial scales. Dittman (2000) measured different size classes of fauna as well as proportions of functional groups and found different patterns of heterogeneity in tropical tidal flats, a characteristic that remains to be tested here. Edgar and Barrett (2002) recorded high variation in density and taxonomic richness within locations of cold temperate tidal flats, while biomass and productivity varied at much larger scales. This illustrates the general importance of multiple ecological measures at multiple scales to describe spatial patterns of fauna. Similar studies in different habitats have included an approach similar to that presented here, for example sub-tidal rocky reefs (Curley et al., 2002), and this approach could easily be adapted to most habitat types. As the representativeness of MPAs should encompass not only all species, but also ecological heterogeneity and diverse habitat function, as broad a suite of ecological measures as possible within logistical and financial limits is desirable.

In this study, heterogeneity of assemblage composition (beta diversity) is greatest at scales of 100s of m, while heterogeneity in abundance and taxonomic richness is equally important at 100s of m as at 10s of km. The proportional contribution to variation within each tidal flat from the different scales may vary between each estuary, but it is the site heterogeneity across 100s of m that stands out in this study. This scale is comparable to scales of heterogeneity in other tidal flats of a similar size (e.g. Dittman, 2000), but as mentioned

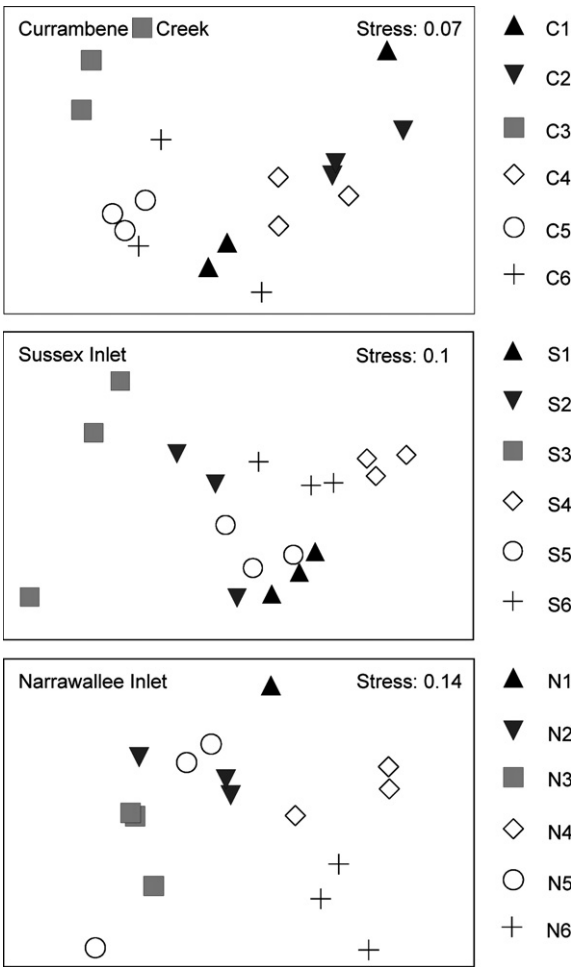

Fig. 4 – MDS plot comparing the composition of macrobenthic assemblages and relative abundances of taxa at sites. Symbols indicate sites 1–6 in each of the three locations. The sites measuring 20 m in diameter are 100s of m apart, while each point represents a 2 m plot of 3 replicate cores.

above, the scale may increase as the area of the tidal flat increases (Rodrigues et al., 2006). Therefore, for conservation purposes it may be more appropriate to refer to heterogeneity at smaller scales as "within location" heterogeneity, as opposed to between location heterogeneity for different estuaries or distinct tidal flats as Edgar and Barrett (2002) do. In terms of composition alone, heterogeneity at this scale might be defined as beta diversity operating within habitat locations. In this case gamma diversity would describe the collective assemblage of taxa of all estuaries as it is the product of alpha diversity (20 m scales) and beta diversity (100s of m) (Schluter and Ricklefs, 1993).

Constable (1999) stated in a review of tidal flat research that there have been few studies that define habitat characteristics, and there is a need to know what size of habitat area is required to maintain the status of an assemblage. Here, we have not addressed the size of a habitat that will optimize resilience or function, but we have provided information on how to achieve the Marine Protected Area goal of representativeness in tidal flats. First, a tidal flat on the South Coast of NSW may represent the gamma diversity within 30 km in terms of dominant taxa, but inclusion of multiple tidal flats would ensure the inclusion of rarer taxa in reserves. Additionally, multiple tidal flats would allow representation of between-location heterogeneity for other ecological characteristics such as abundance or different intensities of within location heterogeneity. Second, whole tidal flat habitat should be protected to encompass the within-location ecological heterogeneity, including beta diversity, which has been identified here as well as in other studies (Dittman, 2000; Edgar and Barrett, 2002; Rodrigues et al., 2006). Ecological heterogeneity is a valuable characteristic of a habitat in its own right. Factors that drive heterogeneity of tidal flat assemblages should be explored further, as degrees of heterogeneity may affect recruitment, function and resilience of these assemblages (Tews et al., 2004). Awareness of these issues will optimize the choices for conservation, as well as set priorities for the future assessment of tidal flats.

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